

CPS I — Exercise Sheet 6

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Exercise 1: Executions, Paths and Traces

- (a)
 - $s_1 \xrightarrow{\alpha} s_1$
 - $s_0 \xrightarrow{\alpha} s_1$
 - $s_1 \xrightarrow{\alpha} s_1 \xrightarrow{\alpha} s_1 \dots$ (infinite)
 - $s_0 \xrightarrow{\alpha} s_1 \xrightarrow{\alpha} s_1 \dots$ (infinite)
- (b)
 - $s_0 \xrightarrow{\alpha} s_1$: Only 1 execution.
 - $s_0 \xrightarrow{\beta} s_2$: At each s_2 , choose β (stay) or γ (go to s_3).

Infinitely many executions.

- (c)
- Infinitely many**
- (d)
 - Trace 1: $\emptyset, \{a\}, \{a\}, \{a\}, \dots$
 - Trace 2 : $\emptyset, \{b\}, \{b\}, \{b\}, \dots$

2 traces .

- (e) Bonus
Yes, it is possible.

Consider a single state s with two distinct self-loops on s : $s \xrightarrow{a} s$ and $s \xrightarrow{b} s$ (where $a \neq b$).

- **Only 1 path**: A path is an infinite sequence of states. Since there's only one state s , the only possible state sequence is $s \rightarrow s \rightarrow s \rightarrow \dots$
- **Infinitely many executions**: At each step, we can choose either transition a or b , leading to infinitely many distinct sequences

Exercise 2: Linear-Time Properties

- (a)

- $T_1 \{A_0 A_1 A_2 \dots \mid A_0 = \{a\} \wedge A_1 = \{b\}\}$
- $T_2 \{A_0 A_1 A_2 \dots \mid \forall i \in \mathbb{N}. (i \text{ odd} \implies a \in A_i) \wedge (i \text{ even} \implies b \in A_i)\}$
- $T_3 \{A_0 A_1 A_2 \dots \mid \forall i \in \mathbb{N}. (A_i = \emptyset \implies (a \in A_{i+1} \vee b \in A_{i+1}))\}$
- $T_4 \{A_0 A_1 A_2 \dots \mid A_0 = \{b\}\}$
- $T_5 \{A_0 A_1 A_2 \dots \mid A_0 = \{a\} \wedge A_1 = \{a\}\} \wedge \{\exists k. \forall i > k. a \notin A_i\}$
- (b)
 - $T_1 \{A_0 A_1 A_2 \dots \mid \forall i \in \mathbb{N}. a \in A_i\}$
 - $T_2 \{A_0 A_1 A_2 \dots \mid \forall i \in \mathbb{N}. a \in A_i\}$
 - $T_3 \{A_0 A_1 A_2 \dots \mid \forall i \in \mathbb{N}. a \in A_i\}$
 - $T_4 \{A_0 A_1 A_2 \dots \mid \forall i \in \mathbb{N}. a \in A_i\}$
 - $T_5 \{A_0 A_1 A_2 \dots \mid \exists i \in \mathbb{N}. a \in A_i \wedge b \in A_i\}$

Exercise 3: Starvation Freedom

- (a) $\text{LIVE}' \subseteq \text{LIVE}$

If a trace $\pi \in \text{LIVE}'$, then for every $i \in \mathbb{N}$:

$$\text{if wait1} \in A_i \text{ then } \exists j \geq i \wedge \text{crit1} \in A_j$$

and similarly for process 2.

If wait1 occurs infinitely often ($\exists^\infty i. \text{wait1} \in A_i$), then infinitely many corresponding j exist such that $\text{crit1} \in A_j$.

Hence:

$$\pi \in \text{LIVE}$$

- (b)

Consider the trace:

$$\pi = \{\text{wait1}\}, \emptyset, \emptyset, \dots$$

- **LIVE:**

wait1 occurs only once, so the antecedent of LIVE is false $\rightarrow$ implication holds $\rightarrow$

$$\pi \in \text{LIVE}$$

- **LIVE':**

At $i = 0$, $\text{wait1} \in A_0$ but there is no $j \geq 0$ with $\text{crit1} \in A_j \rightarrow$

$$\pi \notin \text{LIVE}'$$

- (c) False

In mutual exclusion algorithms with starvation freedom (semaphore or Peterson), every wait is eventually followed by crit.

This means LIVE' automatically holds

- (d) False

The same reason in (a)